

1. 請框出答案. 2. 不可使用手機、計算器，禁止作弊!

1. Find a vector perpendicular to both $[1 - 2i, 2, 2 + i], [i, 1 + i, -3]$.

Answer: $[-7 + 3i, 2 + 4i, 3 + 3i]$

Solution :

Method 1 Gram-Schmitt process

取一不在 $\text{span}(\mathbf{u}, \mathbf{v})$ 的獨立向量（例如標準基底 $\mathbf{e}_1 = [1, 0, 0]$ ），將其對 $\mathbf{u}, \mathbf{v}$ 進行 Gram-Schmidt 正交化，扣除投影量後即可求出與兩者皆正交的第三個向量。

Method 2 Cross product

$$\mathbf{u} \times \mathbf{v} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1 - 2i & 2 & 2 + i \\ i & 1 + i & -3 \end{vmatrix} = \overline{[-7 - 3i, 2 - 4i, 3 - 3i]} = [-7 + 3i, 2 + 4i, 3 + 3i]$$

2. Find value of the dot product $\langle [-3i, 2, -2 + i], [i, 1 + i, -1] \rangle$.

Answer: $1 + 3i$

Solution :

複數內積的定義為 $\langle \mathbf{x}, \mathbf{y} \rangle = \sum \bar{x}_i y_i$ (前項共軛) :

$$(3i)(i) + (2)(1 + i) + (-2 - i)(-1) = 3i^2 + 2 + 2i + 2 + i = 1 + 3i$$

3. Prove that an $n \times n$ matrix U is unitary (i.e. $UU^* = I$) if and only if the rows of U form an orthonormal basis for $\mathbb{C}^n$

Solution :

9-2, problem 35.